

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 3: Shares and Dividend

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Return on investment

1. A company pays a 12% dividend on its shares. An investor's return on his investment is 10%. The shares were bought: (a) at par (b) below par (c) above par (d) cannot be determined **ICSE 2025, 1 mark**

SOLUTION

The return (10%) is less than the dividend rate (12%), so each share cost more than its nominal value.

Answer: (c)

2. A man buys ₹200 shares of a company at a premium of 25%. His return on the investment is 5%. Find (a) the market value of a share (b) the dividend per cent declared (c) the number of shares bought if his annual dividend is ₹1000. **ICSE 2025, 3 marks**

SOLUTION

(a) Market value = $200 + 25\% \text{ of } 200 = 250$. **Market value: ₹250.**

(b) Income per share = $5\% \text{ of } 250 = 12.50$. Dividend per cent = $\frac{12.50}{200} \times 100$. **Dividend: 6.25%.**

(c) Number of shares = $\frac{1000}{12.50}$. **80 shares.**

3. A company has 500 shares of nominal value ₹120 and declares an annual dividend of 15%. Find (a) the total dividend paid by the company (b) the annual income of a man who holds 80 shares (c) the market value of each share if his return is 10%. **ICSE 2020, 3 marks**

SOLUTION

Dividend on one share = $15\% \text{ of } 120 = 18$.

(a) Total dividend = $500 \times 18 = 9000$

(b) Income from 80 shares = $80 \times 18 = 1440$

(c) Return = $\frac{18}{\text{market value}} \times 100 = 10$, so market value = 180.

Answer: (a) ₹9000 (b) ₹1440 (c) ₹180

4. A man invests ₹22,500 in ₹50 shares available at a 10% discount. The company pays a 12% dividend. Find (a) the number of shares bought (b) the annual dividend (c) the rate of return on his investment, to the nearest whole number. **ICSE 2018, 3 marks**

SOLUTION

(a) Market value = $50 - 5 = 45$. Number of shares = $\frac{22500}{45} = 500$.

(b) Dividend on one share = 12% of 50 = 6. Annual dividend = $500 \times 6 = 3000$.

(c) Return = $\frac{3000}{22500} \times 100 = 13.33\%$

Answer: (a) 500 shares (b) ₹3000 (c) about 13%

5. How much must a man invest in ₹50 shares selling at ₹60 to get an income of ₹450, if the dividend is 10%? Find his yield per cent, to the nearest whole number. **ICSE 2017, 3 marks**

SOLUTION

Dividend on one share = 10% of 50 = 5. Number of shares needed = $\frac{450}{5} = 90$.

Investment = $90 \times 60 = 5400$

Yield = $\frac{5}{60} \times 100 = 8.33\%$

Answer: ₹5400; yield about 8%

2 Shares and their values

6. Shares of nominal value ₹75 are available at a discount of 20%. How much is needed to buy 15 shares? (a) ₹60 (b) ₹90 (c) ₹1350 (d) ₹900 **ICSE 2024, 1 mark**

SOLUTION

Market value of one share = $75 - 20\% \text{ of } 75 = 60$

Cost of 15 shares = $15 \times 60 = 900$

Answer: (d)

3 Dividend

7. A man invests ₹33,000 in ₹100 shares of a company at a premium of 10%. The company declares a dividend of 12%. Find (a) the number of shares he buys (b) his annual dividend. **ICSE 2024, 3 marks**

SOLUTION

(a) Market value of one share = $100 + 10 = 110$. Number of shares = $\frac{33000}{110} = 300$.

(b) Dividend on one share = 12% of 100 = 12. Annual dividend = $300 \times 12 = 3600$.

Answer: (a) 300 shares (b) ₹3600

8. A man invests ₹4500 in shares of a company paying a 7.5% dividend. The ₹100 shares are available at a discount of 10%. Find (a) the number of shares he buys (b) his annual income. **ICSE 2019, 3 marks**

SOLUTION

(a) Market value of one share = $100 - 10 = 90$. Number of shares = $\frac{4500}{90} = 50$.

(b) Dividend on one share = 7.5% of 100 = 7.50. Income = $50 \times 7.50 = 375$.

Answer: (a) 50 shares (b) ₹375

4 Selling and reinvesting

9. A man invests ₹8500 in 10%, ₹100 shares at ₹170. He sells them when the price of each share rises by ₹30, and invests the proceeds in 12%, ₹100 shares at ₹125. Find (a) the sale proceeds (b) the number of ₹125 shares he buys (c) the change in his annual income. **ICSE 2019, 4 marks**

SOLUTION

Shares bought first = $\frac{8500}{170} = 50$, with income $50 \times 10 = 500$.

(a) Selling price = $170 + 30 = 200$, so the proceeds are $50 \times 200 = 10000$.

(b) New shares = $\frac{10000}{125} = 80$

(c) New income = $80 \times 12 = 960$. Change = $960 - 500 = 460$.

Answer: (a) ₹10,000 (b) 80 shares (c) income increases by ₹460

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.